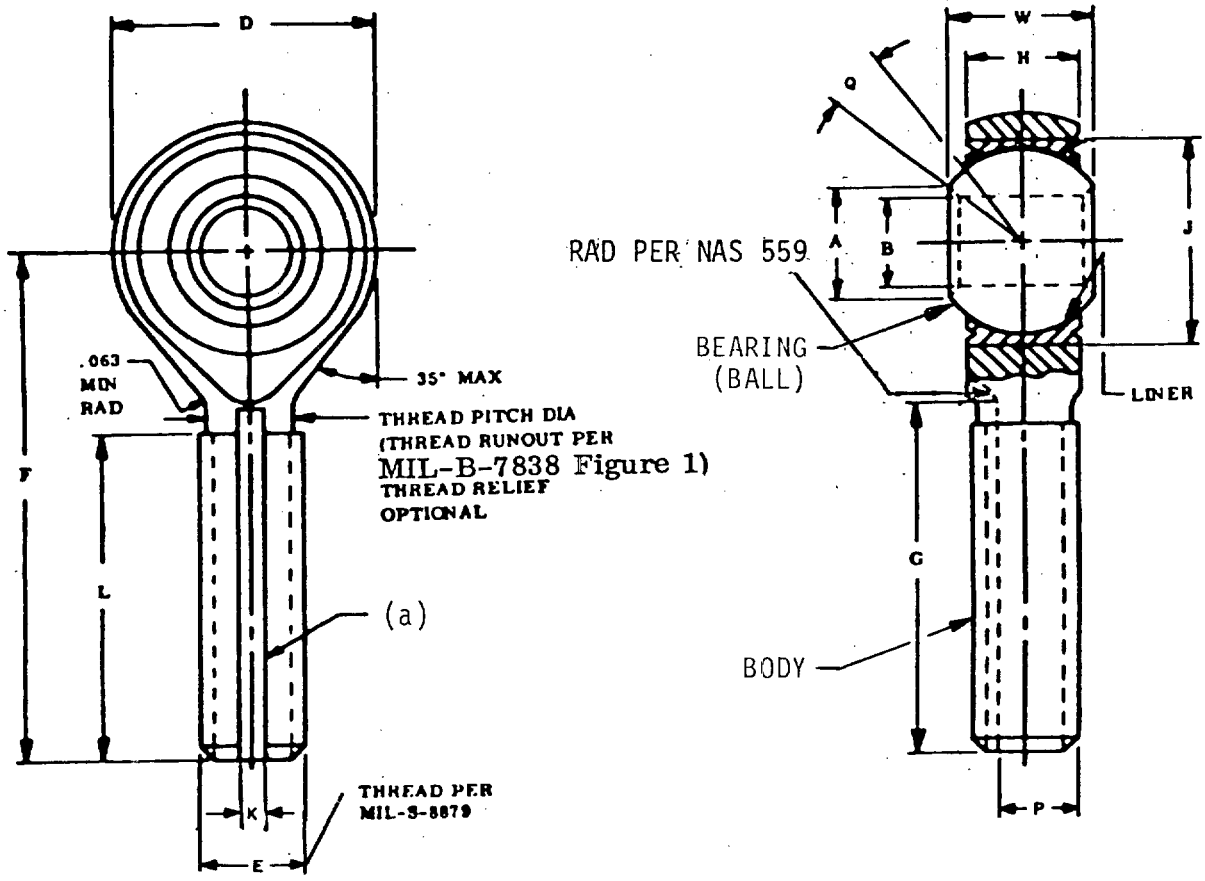
MIL-B-81935/4A
24 JUNE 1983
SUPERSEDING
MIL-B-81935/4
15 February 1982

MILITARY SPECIFICATION SHEET

BEARING, PLAIN, ROD END, SELF-ALIGNING, SELF-LUBRICATING, NARROW, EXTERNALLY THREADED, -65°F TO +325°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the bearings described herein shall consist of this document and the latest issue of specification MIL-B-81935.



(a) KEYWAY SHALL BE ORIENTED ALONG THE CENTERLINE OF THE THREADED SHANK. THE PLANE OF THE BOTTOM OF THE KEYWAY SHALL BE PARALLEL TO THE PLANE MADE BY THE FACE OF THE ROD END HOOP.

FSC 3120

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TABLE I DIMENSIONS

DASH NO.	B BORE +.0000 -.0005	D OUTSIDE DIA ±.010	L THREAD LENGTH ±.031	E THREAD SIZE UNJF-3A	F CTRLINE BALL TO END ±.010	W BALL WIDTH +.000 -.002	H BODY WIDTH ±.005	A MIN	J MAX HOUSING I.D.	Q° MIN	G KEYWAY FLAT +.000 -.020	K KEYWAY (a) +.005 -.000	P KEYWAY (a) -.000 -.005
-3	.1900	.680	.775	.2500-28	1.315	.281	.228	.293	.5625	10	.896	.062	.207
-4	.2500	.827			1.443	.343	.260	.364	.6562				
-5	.3125	.984	1.187	.3125-24	1.948	.375	.291	.419	.7500				.268
-6	.3750	1.131		.3750-24	2.030	.406	.322	.475	.8125	9	1.308		.319
-7	.4375	1.294	1.281	.4375-20	2.250	.437	.353	.530	.9062		1.402	.093	.383
-8	.5000	1.459	1.468	.5000-20	2.544	.500	.400	.600	1.0000		1.589		.445
-10	.6250	1.763	1.562	.6250-18	2.832	.625	.510	.739	1.1875	8	1.683	.125	.541
-12	.7500	2.140	1.687	.7500-16	3.193	.750	.603	.920	1.4375		1.808		.663
-14	.8750	2.372	2.000	.8750-14	3.677	.875	.713	.980	1.5625		2.121	.156	.777
-16	1.0000	2.681	2.100	1.0000-12	3.968	1.000	.807	1.118	1.7500	9	2.221		.900

(a) Keyway, when specified, is compatible with locking devices MIL-B-81935/3 for sizes -3 thru -8, and NAS 513 for sizes -10 thru -16. Keyway tolerances not specified shall be in accordance with MIL-B-81935/3 or NAS 513 as applicable.

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TABLE II LOAD VALUES

DASH NO.	1/ ULTIMATE STATIC LOAD (LBS)	2/ FATIGUE LOAD (LBS)	AXIAL PROOF LOAD (LBS)	WEIGHT MAX (LBS)	NO LOAD ROTATIONAL BREAKAWAY TORQUE (IN - LBS)	
					MIN	MAX
-3	3,000	1,100	150	.038	.5	6
-4	5,300	1,500	430	.045		
-5	8,600	2,400	700	.081	1	15
-6	13,000	3,600	1,100	.120		
-7	17,800	5,000	1,400	.172		
-8	24,200	6,800	2,040	.254		
-10	38,500	10,800	2,430	.455		
-12	56,600	16,000	2,940	.774		
-14	77,400	21,900	3,190	1.141	1	24
-16	101,400	28,600	3,570	1.646		

1/ ULTIMATE LOADS ARE ANALYTICAL VALUES BASED ON ROD END BANJO FOR -3 AND THREADS FOR ALL OTHER SIZES.

2/ FATIGUE LOADS ARE ANALYTICAL VALUES BASED ON 50,000 CYCLE LIFE. FATIGUE LOADS DEFINED PER MIL-B-81935, SEC 4.6.3.

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REQUIREMENTS:

MATERIAL:	
BODY:	ALLOY STEEL PER MIL-S-5000 (4340)
BEARING:	PER MIL-B-81820 AND MS14101
HARDNESS:	HEAT TREATMENT IN ACCORDANCE WITH MIL-H-6875
BODY:	Rc 39-42
SURFACE TEXTURE:	PER ANSI B46.1
BODY:	BORE: Ra 32 MAX SIDES OF THREAD AND ROOT AREA: Ra 32 THREAD RELIEF: Ra 63 ALL OTHER MACHINED SURFACES: Ra 125 MAX
PLATING:	CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2
TOLERANCES:	UNLESS OTHERWISE SPECIFIED: DECIMALS ± .010; ANGLES ± 1/2°

DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

BREAK SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

PART NUMBER: M81935/4 XX X X

- X — ADD L FOR LEFT HAND THREAD WHEN SPECIFIED
- X — KEYWAY (CODE K) WHEN SPECIFIED
- XX — BORE DIAMETER CODE IN MULTIPLES OF 1/16 INCH
- M81935/4 — BASIC PART NUMBER

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army -
Air Force - 99
DLA - IS

Preparing activity:
NAVY - AS

(Project No. 3120-0612)

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